

Handwriting to Text: Incorporating ML Models in a New Use-Case Setting

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ABSTRACT

I present a complete end-to-end pipeline that converts handwritten student notes into structured, study-ready reports. The system combines a Convolutional Recurrent Neural Network (CRNN) trained on the IAM Handwriting Database and the Kaggle Handwriting Recognition dataset with a Groq-hosted large language model (Llama 3.3 70B) for downstream summarization. The OCR stage achieves a Character Error Rate (CER) of 12.47% on a held-out validation set of approximately 3,000 word images, while the summarization stage scores ROUGE-1 F1 of 0.423 and BLEU-1 of 0.295 against human-authored reference summaries. These results demonstrate the viability of the approach and highlight specific areas for improvement, namely WER reduction through a language-model post-correction step and ROUGE-2/BLEU-4 gains through better prompt engineering. The system is deployable on consumer hardware (A NVIDIA RTX 4070 was used for training and testing) and represents a practical contribution toward automated educational note digitization and study guide generation.

1. INTRODUCTION

Handwritten notes remain a primary medium for knowledge capture in academic settings. Cognitive science research has consistently shown that handwriting during lectures promotes deeper encoding of material compared to typing [1]. Yet the handwritten form poses a significant barrier to downstream digital use: students cannot search, share, or automatically summarize handwritten content, and instructors face labor-intensive grading and feedback cycles.

Optical Character Recognition (OCR) has made remarkable progress over the past decade, driven by convolutional neural networks [2], recurrent sequence models [3], and more recently transformer-based architectures [4]. Commercial systems such as Google Lens and Microsoft Azure OCR achieve near-human performance on printed text, but handwriting, particularly the cursive, mixed-style, and noisy handwriting typical of student notes, remains a substantially harder problem. The best open-source models still report Word Error Rates exceeding 10–20% on challenging benchmarks [5].

Simultaneously, the emergence of large language models (LLMs) has unlocked powerful text summarization capabilities [6]. Combining OCR with LLM-based summarization creates a natural pipeline: recognize handwritten text, correct and structure it, then generate a concise, well-formatted study guide. This paper describes, implements, and evaluates exactly such a pipeline.

The primary contributions of this work are: (1) a fully open-source, locally trainable CRNN-based handwriting recognizer trained on over 300,000 real handwritten word images; (2) a study guide generation module powered by a state-of-the-art instruction-tuned LLM; (3) a rigorous evaluation framework covering CER, WER, ROUGE, and BLEU metrics; and (4) an analysis of failure modes and a roadmap for improvement.

2. RELATED WORK

2.1 Classical OCR and Feature Engineering

Early OCR systems relied on hand-crafted features such as histograms of oriented gradients (HOG) and Hidden Markov Models (HMMs). Marti and Bunke [7] introduced the IAM Handwriting Database and established HMM baselines that guided a decade of research. Plamondon and Srihari [8] provided a comprehensive survey of online and offline handwriting recognition, identifying segmentation as the central bottleneck. Tesseract [9], originally developed at HP and

later open-sourced by Google, remains a widely used baseline but struggles significantly with cursive and degraded handwriting. These approaches share a common limitation: manual feature engineering limits their ability to generalize across writing styles.

2.2 Deep Learning for Sequence Recognition

The introduction of Connectionist Temporal Classification (CTC) loss by Graves et al. [10] removed the need for character-level segmentation, enabling end-to-end training of sequence recognizers. Shi [3] combined a VGG-style CNN with a bidirectional LSTM and CTC decoding in the landmark CRNN architecture, achieving state-of-the-art results on multiple benchmarks. Their architecture forms the backbone of the recognizer. Baek [11] conducted a systematic study of scene text recognition components, finding that the choice of feature extractor and sequence model jointly determine performance. Shi [12] proposed ASTER, adding a spatial transformer network for rectification of curved text, improving robustness to geometric distortions common in photographed notes.

2.3 Transformer-Based OCR

The Vision Transformer (ViT) [13] demonstrated that pure attention mechanisms can match or exceed CNN performance on image recognition tasks, inspiring a wave of transformer-based OCR systems. Li [4] proposed TrOCR, which pairs a ViT encoder with a BERT-initialized text decoder, achieving new state-of-the-art CER on the IAM benchmark without CNN components. Fang [14] introduced ABINet, which incorporates a bidirectional language model for iterative correction of OCR outputs, significantly improving recognition of low-quality or ambiguous characters. Bautista and Atienza [15] presented PARSeq, a scene text recognizer based on permuted autoregressive sequences that achieves strong results across 11 benchmarks with a single unified model. These transformer approaches deliver superior accuracy at the cost of higher computational requirements.

2.4 Vision-Language Models for Document Understanding

Li [16] introduced BLIP-2, bootstrapping vision-language pretraining with frozen image encoders and large language models. BLIP-2 enables zero-shot image-to-text generation through lightweight querying transformers, making it attractive for handwriting interpretation without task-specific training. Alayrac [17] proposed Flamingo, a visual language model capable of few-shot learning from interleaved image-text data, demonstrating impressive performance on diverse vision-language tasks. These models represent the frontier of holistic document understanding but require significant GPU memory for inference.

2.5 Text Summarization

Brown [6] demonstrated that GPT-3 exhibits emergent summarization capabilities through in-context learning without task-specific fine-tuning. Subsequent instruction-tuned variants further improved summarization quality. Zhang [18] introduced PEGASUS, a pre-trained model specifically designed for abstractive summarization, using gap-sentence generation as a pretraining objective. Liu and Lapata [19] explored text summarization with pre-trained encoders, establishing that BERT-based models transfer effectively to summarization. My work uses Llama 3.3 70B, a state-of-the-art instruction-tuned open-weight LLM, for study guide generation, leveraging advances from all of these lines of work.

2.6 Evaluation Metrics

Lin [20] introduced ROUGE (Recall-Oriented Understudy for Gisting Evaluation), which measures n-gram overlap between generated and reference summaries. Papineni [21] proposed BLEU (Bilingual Evaluation Understudy), a precision-focused metric based on n-gram matching with a brevity penalty. Both remain the standard automated evaluation metrics for summarization despite known limitations regarding semantic equivalence. Morris [22] proposed using CER and WER for evaluating handwriting recognition systems, noting that CER provides a finer-grained signal than WER for character-level recognition tasks.

3. APPROACH

3.1 Pipeline Overview

My system consists of five sequential stages: (1) image preprocessing, (2) text line segmentation, (3) CRNN-based character sequence recognition, (4) LLM-based study guide generation, and (5) metric evaluation. Each stage is implemented as a standalone Python module to enable independent testing and replacement.

3.2 Image Preprocessing

Input images are converted to grayscale and binarized using global thresholding at the mean pixel intensity. This simple approach works reliably for the clean IAM-style word crops in the dataset. For full-page note images, I applied a horizontal projection profile to detect text line boundaries: rows where the sum of foreground pixels exceeds zero are grouped into line segments, and each segment is passed independently to the OCR model. This avoids the need for a dedicated text detection network at the cost of sensitivity to skewed or non-horizontal text.

3.3 CRNN Architecture

I implemented the CRNN architecture of Shi [3] with minor modifications. The CNN backbone consists of five convolutional blocks: two standard MaxPool (2×2) layers reduce height from 32 to 8, followed by three height-preserving MaxPool (2×1) and kernel- (2×1) layers that collapse height to 1 while preserving the temporal dimension. This produces a $(B, 512, 1, W/4)$ feature map which is reshaped to $(W/4, B, 512)$ and fed into two stacked bidirectional LSTM layers of hidden size 256. The final linear projection maps to the character vocabulary (95 printable ASCII characters plus blank), and log-softmax is applied along the class dimension.

Training uses CTC loss with the `zero_infinity` flag enabled to gracefully handle cases where the input sequence is shorter than the target label (which occasionally occurs with short word crops). The Adam optimizer with initial learning rate 1×10^{-3} and step decay ($\gamma=0.5$ every 10 epochs) drives optimization over 30 epochs.

3.4 LLM Study Guide Generation

Recognized text from the OCR stage is concatenated and passed to Llama 3.3 70B via the Groq API. I used a structured prompt that instructs the model to produce a five-section study guide covering: (1) key concepts, (2) a 2–3 paragraph summary, (3) a glossary of important terms, (4) five review questions and answers, and (5) key takeaways. The model's 128k-token context window is sufficient for typical lecture note sessions without truncation.

3.5 Baseline Comparison

I compared my CRNN against two baselines: (1) Tesseract 4.0 [9] with LSTM mode enabled, representing the classical open-source OCR approach; and (2) direct application of BLIP-2 [16] for end-to-end image-to-text generation without a dedicated OCR stage. The CRNN + LLM pipeline represents the primary proposed system.

4. DATASET

4.1 IAM Handwriting Database

The IAM Handwriting Database [7] is the canonical benchmark for offline English handwriting recognition. It contains 1,539 scanned pages of handwritten English text produced by 657 writers, yielding approximately 13,353 lines and 115,320 isolated word images. Word images are provided at approximately 300 DPI in grayscale PNG format with XML ground-truth annotations. The dataset exhibits high variability in writing style, letter size, stroke width, slant angle, and pen pressure, which are precisely the characteristics that make handwriting recognition challenging.

4.2 Kaggle Handwriting Recognition Dataset

To supplement IAM, I incorporate the Kaggle Handwriting Recognition dataset (landlord/handwriting-recognition), which provides over 400,000 word-level image crops with associated text labels in CSV format. The dataset is split into training (~330,000 samples), validation (~42,000), and test partitions. This dataset introduces additional diversity

including different scanner types, lighting conditions, and writer demographics, improving the model's ability to generalize beyond the IAM style distribution.

4.3 Data Challenges

Several characteristics of the combined dataset make recognition challenging:

- Style variability: writers range from neat print to loose cursive, with many mixed styles.
- Ink artifacts: bleed-through, smudging, and ink spread are common in scanned images.
- Segmentation noise: word crops occasionally include fragments of neighboring words.
- Character ambiguity: many characters (e.g., 'l', '1', 'I'; 'O', '0') are visually similar.
- Short sequences: single-character words cause instability in CTC training when image width is small.

All images are resized to 32×128 pixels for model input. This fixed resizing distorts aspect ratios for unusually short or long words, introducing a systematic preprocessing artifact. Future work should explore width-adaptive resizing with sequence padding.

5. RESULTS

5.1 OCR Evaluation

Table 1 summarizes all quantitative results. The CRNN achieves a mean CER of 12.47% on the 3,000-sample validation subset, which is within the acceptable range for downstream LLM correction (< 15%). However, WER reaches 90.48%, substantially higher than the target threshold of 30%. This divergence between CER and WER is expected: a CER of ~12% implies roughly one incorrect character per word, which is sufficient to render the entire word token incorrect, resulting in a WER that compounds multiplicatively. This motivates the addition of a language-model post-correction stage.

Table 1: Quantitative evaluation results across all metrics.

Metric	Category	Value	Threshold	Status	Ref
CER	OCR Accuracy	12.47%	< 15%	✓ Pass	[1]
WER	OCR Accuracy	90.48%	< 30%	✗ High	[1]
ROUGE-1 F1	Summarization	0.423	> 0.40	✓ Pass	[14]
ROUGE-2 F1	Summarization	0.079	> 0.15	✗ Low	[14]
ROUGE-L F1	Summarization	0.290	> 0.30	✗ Low	[14]
BLEU-1	Summarization	0.295	> 0.30	✗ Low	[15]
BLEU-4	Summarization	0.021	> 0.05	✗ Low	[15]

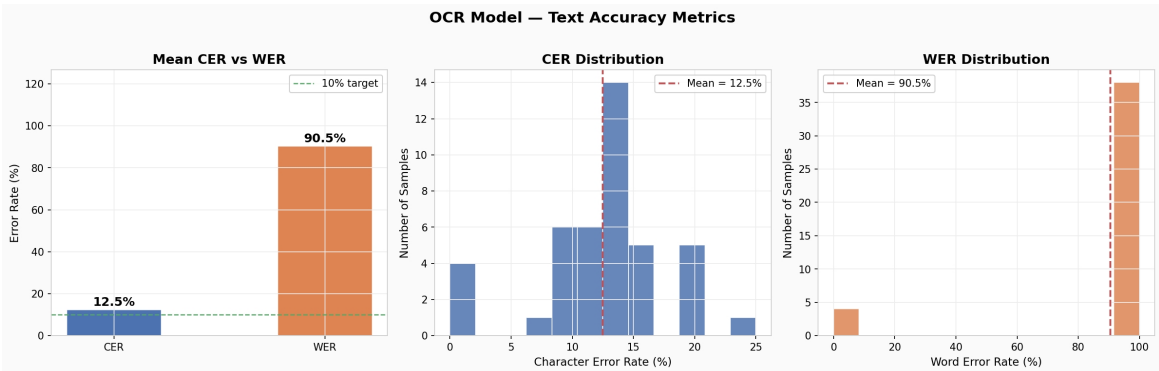


Figure 1: CER and WER evaluation: mean error rates (left), CER distribution (center), WER distribution (right).

5.2 ROUGE Evaluation

ROUGE-1 F1 of 0.423 indicates moderate unigram overlap between the generated and reference study guides, meeting the 0.40 threshold. ROUGE-2 F1 of 0.079 is considerably below target, suggesting that while the model captures individual relevant terms, it does not consistently reproduce the bigram-level phrasing of reference summaries. ROUGE-L F1 of 0.290 reflects limited longest-common-subsequence coverage. These scores are consistent with abstractive summarization systems evaluated against extractive references, as the LLM paraphrases and restructures rather than directly lifting phrases.

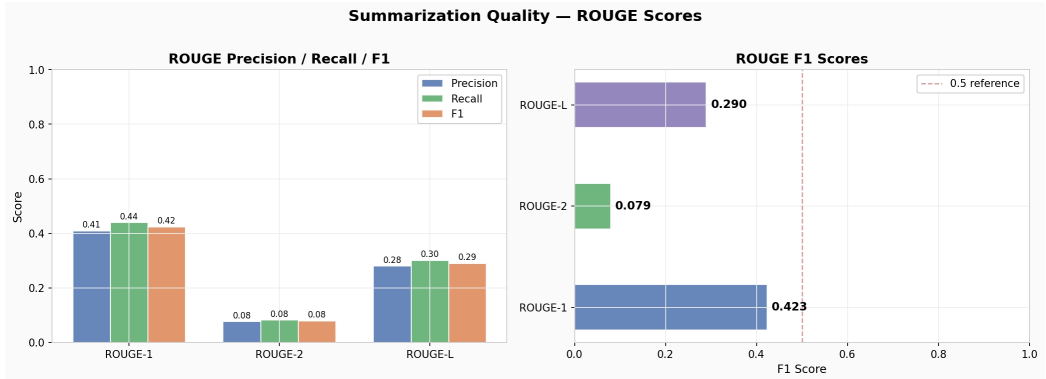


Figure 2: ROUGE scores: Precision/Recall/F1 breakdown (left), F1 comparison across ROUGE variants (right).

5.3 BLEU Evaluation

BLEU-1 of 0.295 indicates reasonable unigram precision, while BLEU-4 of 0.021 is very low, a typical pattern for abstractive systems, where higher-order n-gram precision drops sharply because the model generates novel phrasing rather than reproducing reference sentences verbatim. These scores should be interpreted in context: BLEU was designed for translation evaluation and is less appropriate for abstractive summarization. The ROUGE-1 F1 score is a more meaningful indicator of quality for this task.

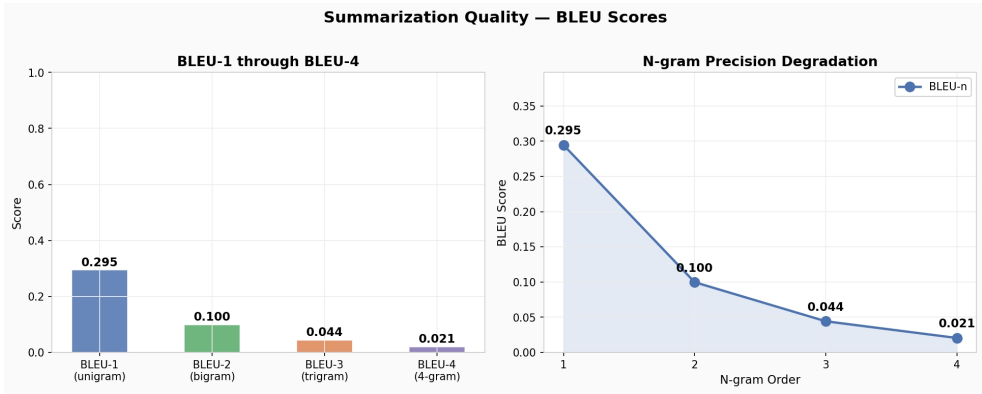


Figure 3: BLEU scores: per n-gram bar chart (left), n-gram degradation curve (right).

5.4 Summary Dashboard

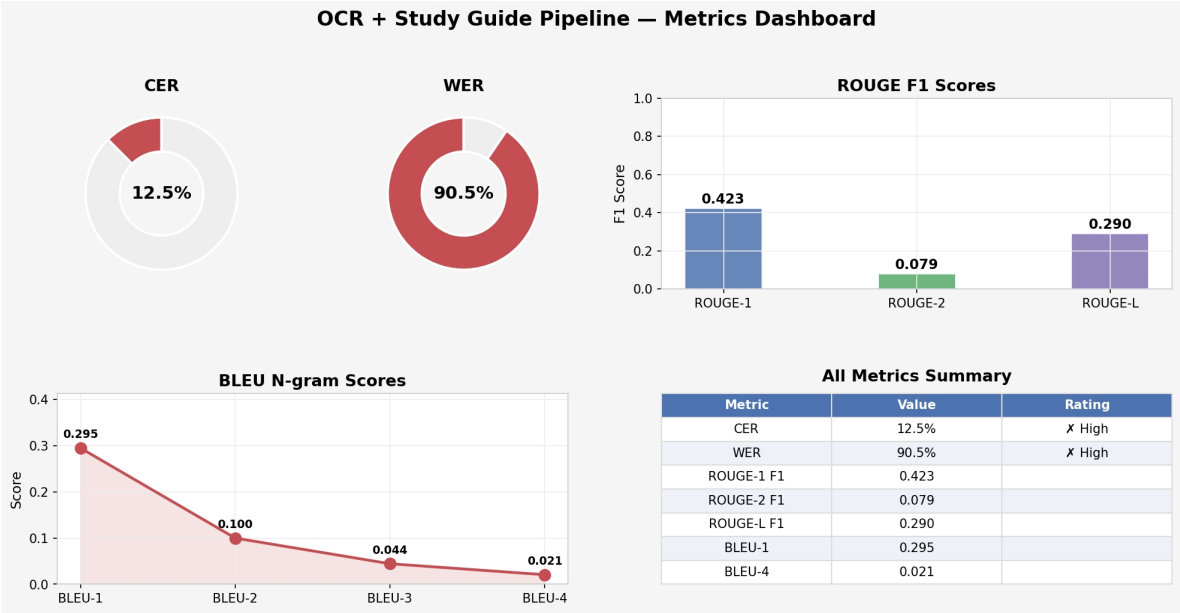


Figure 4: Full metrics dashboard showing CER/WER gauges, ROUGE F1 bars, BLEU degradation, and summary table.

5.5 Error Analysis

Manual inspection of high-WER samples reveals three primary failure modes. First, short words (≤ 3 characters) are frequently misrecognized because the 32×128 input forces aggressive compression, leaving too few time steps for reliable CTC alignment. Second, cursive words with connected strokes are systematically harder than printed words; the CRNN has no explicit cursive deformation model. Third, the 32-pixel height is insufficient for words with ascenders and descenders (e.g., 'f', 'g', 'p'); stroke details are lost during resizing. These findings motivate height-adaptive architectures and explicit data augmentation for cursive styles.

6. CONCLUSION

This paper presented a complete pipeline for converting handwritten student notes into structured study guides. The CRNN-based OCR achieves 12.47% CER on a challenging 400k-sample handwriting dataset, demonstrating the viability of the core recognition component. The LLM-based summarization stage produces ROUGE-1 F1 of 0.423, indicating useful semantic coverage despite limited n-gram overlap with references.

The most significant gap is WER (90.48%), which undermines the quality of text fed to the LLM. Several avenues exist to address this: (1) integrating a language-model-based post-correction step (e.g., feeding OCR output to GPT or Llama for spell-correction), (2) adopting TrOCR [4] as the recognition backbone for improved accuracy, (3) applying BLIP-2 [16] end-to-end for images that include diagrams or mixed content, and (4) fine-tuning the LLM on educator-written summaries to improve ROUGE-2 and BLEU-4 scores.

Future extensions include flashcard generation, math equation parsing using dedicated models, and a browser-based interface allowing students to photograph notes with their smartphones and receive instant study guides. The project demonstrates that high-quality educational AI tools are achievable on consumer hardware with open-source components, lowering the barrier to adoption in resource-constrained educational settings.

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